

THE KLEENE INTERPOLATION: BOUNDARY FIBERS AND THE STRICT LAYER ACROSS THE SUBVARIETIES OF DE MORGAN ALGEBRAS

RYOTA SHIBUSAWA

ABSTRACT. Two companion papers determined the boundary fibers of free De Morgan algebras (Boolean intervals 2^D over antichains of diagonal points, Sperner-extremal) and the resulting structure of the strict layer of De Morgan cubical sets (a defect formula and a coherence dichotomy). This paper computes the same theory over the remaining subvarieties of De Morgan algebras — by Kalman’s classical theorem, exactly the Boolean and Kleene varieties. The free Kleene algebra on n generators is identified with the lattice of monotone Boolean functions on the *unmixed* part of the literal cube, the points not containing both a $(0, 0)$ and a $(1, 1)$ pair; the identification recovers the free spectra $6, 84, 43918, \dots$ of Berman–Mukaidono. Only two unmixed points are invisible to the faces — the poles — and they are comparable, so *boundary fibers in free Kleene algebras have at most two elements, with exactly two non-singleton fibers in every dimension*; realizable boundaries therefore number two fewer than elements, matching Berman’s constant-free spectrum (OEIS A007154) and giving that sequence a new, cubical interpretation. The defect formula and the coherence dichotomy port verbatim, with defects confined to the poles. The result is a complete classification of strict-layer behaviour across the subvariety lattice: Boolean and bare distributive interval theories have singleton fibers (Eilenberg–Zilber-compatible degeneracy behaviour), Kleene has exactly two polar exceptions independent of dimension, and De Morgan is Sperner-extremal — the three grades of involution measure the distance from Eilenberg–Zilber structure. A completeness theorem closes the classification along the signature axis as well: nontrivial boundary fibers require both the involution and a connection in the interval signature, and once both are present De Morgan’s laws and Kalman’s theorem leave no room beyond the three grades. All counts and theorems are verified exhaustively by machine through three generators.

1. INTRODUCTION

The interval theories of cubical type theory are bounded distributive lattices with extra structure, and the free algebras of those theories govern the *strict* (judgmental) layer of the resulting cubical sets: their elements are the

2020 *Mathematics Subject Classification*. 06D30; 06A07, 03B38, 55U35.

Key words and phrases. Kleene algebra, De Morgan algebra, free algebra, cubical sets, boundary fiber, free spectrum, Eilenberg–Zilber category, cubical type theory.

reparametrizations a proof assistant applies silently. Over the De Morgan interval theory of Cohen–Coquand–Huber–Mörtberg [5], two companion papers analyzed this layer. In [8] the *boundary fibers* of the free De Morgan algebra DM_n — the sets of elements sharing all $2n$ face restrictions — were shown to be Boolean intervals 2^D , with D an antichain of *diagonal* points of the literal cube, of size up to $2^{\binom{n}{\lfloor n/2 \rfloor}}$ by Sperner’s theorem; in [9] this non-uniqueness was organized into a structure theory for the De Morgan cube category (a one-step *defect formula* and an exact *coherence dichotomy*), against the background fact that the category is not Eilenberg–Zilber.

By a classical theorem of Kalman [7], the variety of De Morgan algebras has exactly two proper nontrivial subvarieties: Boolean algebras and *Kleene algebras* (De Morgan algebras satisfying $x \wedge \neg x \leq y \vee \neg y$) — the three-element chain **3** being the generating subdirectly irreducible of the latter. Each subvariety has its cube category and its cubical sets [4], and it is natural to ask how the fiber and strict-layer theory varies across the lattice of subvarieties. This paper answers the question completely; the summary is the following table, in which the last row is [8, 9] and the first is folklore.

theory	invisible pts	fibers	non-singleton fibers
Dedekind, Boolean	none	1	none
Kleene	the two poles	≤ 2	exactly 2, every n
De Morgan	2^n diagonals	$\leq 2^{\binom{n}{\lfloor n/2 \rfloor}}$	34.5% of DM_3

The Kleene row is the content of this paper, and it sits strictly — and, as it turns out, very asymmetrically — between its neighbours: the Kleene law does not restore unique fillers, but it cuts the exceptional locus from a positive fraction of all elements down to *exactly two fibers, independent of dimension*.

The technical basis is a representation theorem (Theorem 2.2) that seems worth recording in its own right. Call a point of the literal cube $\{0, 1\}^{2n}$ *mixed* if some coordinate pair equals $(0, 0)$ and another equals $(1, 1)$, and let U_n be the subposet of unmixed points. Then the free Kleene algebra KL_n is the lattice of monotone Boolean functions on U_n — the Kleene law deletes from the De Morgan representation [8] exactly the mixed points. The proof is a two-sided probe of the three-valued semantics: the lower and upper Boolean approximations of a point of $\mathbf{3}^n$ are always unmixed, and every unmixed point arises as such an approximation. The cardinalities $|\text{KL}_n| = 6, 84, 43918, \dots$ then agree with the free spectra computed by Berman–Mukaidono [2] (OEIS A268534), which we take as independent confirmation.

The fiber theory follows in one stroke (Theorem 3.2): the points of U_n invisible to all faces are the two poles $\perp, \top$ — every other diagonal point of the De Morgan literal cube is mixed — and the poles are comparable, so the defect antichains of [8, 9] can only be $\emptyset$, $\{\perp\}$ or $\{\top\}$. Fibers have at most two elements; exactly two fibers are non-singleton (the *polar* fibers $\{0, \delta_\top\}$ and $\{1 \setminus \perp, 1\}$); and the number of realizable boundaries is exactly

$|\text{KL}_n| - 2$. Numerically this is the sequence $4, 82, 43916, \dots$ — which turns out to be in the OEIS as A007154, *Berman’s constant-free free spectrum of Kleene algebras* [1]; the coincidence is explained by a two-line bijection (boundaries $\leftrightarrow$ fiber-minimal elements, of which all but two elements are), giving the classical sequence a new interpretation as a count of spheres in the Kleene cube category.

Finally, the strict-layer theory of [9] ports verbatim (§4): the Kleene cube category is not idempotent complete (the proof for De Morgan applies word for word, as $|\text{KL}_1| = |\text{DM}_1| = 6$), its Karoubi envelope is the opposite of finite projective Kleene algebras via Bova–Cabrer [3, Thm. 12], the defect formula holds with defects confined to the poles, and the coherence dichotomy is exact — with the exceptional locus of any substitution now consisting of at most the two polar elements. All statements are verified exhaustively by machine for $n \leq 3$ (§6).

Reading guide. This paper is the third of a series but is self-contained in its statements; proofs that are verbatim ports from [8, 9] are indicated as such, with the one substitution “diagonal points $\mapsto$ poles” spelled out. The conceptual through-line is stated once, here: the lattice of subvarieties of De Morgan algebras (trivial $\subset$ Boolean $\subset$ Kleene $\subset$ De Morgan, by Kalman) is simultaneously a scale of *how badly the associated cube category fails to be Eilenberg–Zilber*, with the failure measured by boundary-fiber structure — none, two polar bits, a full Sperner antichain of bits. Section 5 shows the scale is complete: no interval signature produces any other fiber behaviour.

2. THE REPRESENTATION THEOREM

A *Kleene algebra* is a De Morgan algebra satisfying $x \wedge \neg x \leq y \vee \neg y$; the variety is generated by the three-element chain $\mathbf{3} = \{0, u, 1\}$ with $\neg u = u$ [7, 6]. Write KL_n for the free Kleene algebra on $x_1, \dots, x_n$. We use the literal-cube representation of DM_n from [8]: monotone functions (up-sets) on $Q_n = \{0, 1\}^{2n}$, coordinates in pairs $(x_i, \neg x_i)$.

Definition 2.1. A point $\alpha \in Q_n$ is *mixed* if some pair $\alpha_i = (0, 0)$ and some pair $\alpha_j = (1, 1)$; otherwise *unmixed*. Let $U_n \subseteq Q_n$ be the subposet of unmixed points. ($|U_1| = 4$, $|U_2| = 14$, $|U_3| = 46$.)

Note that U_n is closed under the De Morgan symmetry (complement all coordinates, swap each pair), and that the *diagonal* points of Q_n (all pairs in $\{(0, 0), (1, 1)\}$) that are unmixed are exactly the two poles $\perp = (0, \dots, 0)$ and $\top = (1, \dots, 1)$.

Theorem 2.2. *Restriction of monotone functions along $U_n \hookrightarrow Q_n$ induces an isomorphism from the free Kleene algebra onto the lattice of monotone functions on U_n , with involution $(\neg\varphi)(\alpha) = 1 - \varphi(\sigma\bar{\alpha})$ as before:*

$$\text{KL}_n \cong \{\text{monotone } U_n \rightarrow \{0, 1\}\}.$$

Proof. The restriction map $\rho : \text{DM}_n \rightarrow \text{Mon}(U_n)$ is a homomorphism of De Morgan algebras: meets, joins and bounds are pointwise, and U_n is closed

under the symmetry defining $\neg$. It is surjective, since a monotone function on U_n extends to Q_n (e.g. by its least monotone extension).

$\text{Mon}(U_n)$ is a Kleene algebra: if $\alpha \in x_i \wedge \neg x_i$ then $\alpha_i = (1, 1)$, so by unmixedness no pair of α is $(0, 0)$; hence $\alpha_j \in \{(0, 1), (1, 0), (1, 1)\}$ and $\alpha \in x_j \vee \neg x_j$, for every j . Thus ρ factors through the Kleene quotient, giving a surjection $\text{KL}_n \twoheadrightarrow \text{Mon}(U_n)$.

For injectivity we use the three-valued semantics. By [6], KL_n embeds into $\mathbf{3}^{(3^n)}$ as the subalgebra generated by the projections; write $\hat{\varphi} : \mathbf{3}^n \rightarrow \mathbf{3}$ for the function induced by φ . For $v \in \mathbf{3}^n$ define $v_{\text{low}}, v_{\text{high}} \in Q_n$ by encoding $v_i = 0 \mapsto (0, 1)$, $v_i = 1 \mapsto (1, 0)$, and $v_i = u \mapsto (0, 0)$ (low), resp. $(1, 1)$ (high). Both are *unmixed*, because the choice for u is made uniformly; and a direct induction on terms shows $\hat{\varphi}(v) = 0, u, 1$ according as the pair $(\varphi(v_{\text{low}}), \varphi(v_{\text{high}}))$ is $(0, 0)$, $(0, 1)$, $(1, 1)$ (monotonicity excludes $(1, 0)$). Hence $\hat{\varphi}$ is determined by $\rho(\varphi)$. Conversely every unmixed point is a v_{low} or a v_{high} (read u at its diagonal pairs), so $\rho(\varphi)$ is determined by $\hat{\varphi}$; as $\varphi \mapsto \hat{\varphi}$ is injective on KL_n , so is the induced map $\text{KL}_n \rightarrow \text{Mon}(U_n)$. $\square$

Remark 2.3. The counts $|\text{Mon}(U_n)| = 6, 84, 43918$ for $n = 1, 2, 3$ (machine enumeration, §6) agree with the free spectra of Kleene algebras computed by Berman–Mukaidono [2] (OEIS A268534), an independent confirmation of Theorem 2.2. The theorem refines the classical normal-form descriptions [6, 2] in the same way the literal-cube picture refines them for De Morgan algebras: it makes the *boundary* combinatorics visible. In that picture the Kleene law acts by *deleting the mixed points* — and, as the next section shows, mixed points are almost all of the diagonal.

3. BOUNDARY FIBERS: AT MOST TWO ELEMENTS

Faces $\partial_i^e : \text{KL}_n \rightarrow \text{KL}_{n-1}$ restrict to the face subcubes of U_n exactly as in the De Morgan case (the face subcubes of U_n are copies of U_{n-1} : fixing a pair to $(0, 1)$ or $(1, 0)$ cannot create or destroy mixedness). A point of U_n is a *face point* if some pair is $(0, 1)$ or $(1, 0)$; the points invisible to every face are the unmixed diagonal points:

Lemma 3.1. *The unmixed diagonal points of Q_n are exactly the poles $\perp$ and $\top$; they are comparable, and there is a face point strictly between them (any point with one pair $(0, 1)$ and the rest $(0, 0)$).*

Theorem 3.2. *Every boundary fiber of KL_n is an interval $[m, M]$ with $D = M \setminus m$ an antichain contained in $\{\perp, \top\}$; hence $|D| \leq 1$ and fibers have at most two elements. Exactly two fibers are non-singleton in every dimension $n \geq 1$: the zero-boundary fiber $\{0, \delta_\top\}$ ($\delta_\top$ the indicator of $\top$, i.e. $x_1 \wedge \neg x_1 \wedge \cdots \wedge x_n \wedge \neg x_n$) and the top-boundary fiber $\{1 \setminus \perp, 1\}$. Consequently*

$$B_{\text{KL}}(n) = |\text{KL}_n| - 2,$$

where $B_{\text{KL}}(n)$ is the number of realizable boundaries.

Proof. The interval property, the confinement of D to invisible points, and the antichain property are the proofs of [8, §3] verbatim, applied to the poset U_n : the only inputs are that faces see exactly the face points and that between comparable invisible points there is a face point (Lemma 3.1). Since $\perp < \top$, an antichain contains at most one pole. Realizability of $D = \{\top\}$ and $D = \{\perp\}$ is witnessed by the displayed fibers (that $\delta_\top$ is monotone on U_n and has zero boundary, and dually, is immediate); uniqueness of each — i.e. that no other fiber has $D \neq \emptyset$ — follows from the fiber description of [8, Thm. 3.4]: $D = \{\top\}$ forces every face point above $\top$ to lie in m , and there are none, so m is the least up-set with empty intersection with $\{\top\}$'s up-set, namely $m = 0$; dually for $\perp$, m must contain every face point, forcing $m = 1 \setminus \perp$. Finally, boundaries biject with fiber-minimal elements [8, Thm. 4.2], and all elements are fiber-minimal except the top members $\delta_\top$ and 1 of the two polar fibers. $\square$

Remark 3.3 (the free spectrum, interpreted). The sequence $B_{\text{KL}}(n) = 4, 82, 43916, \dots$ appears in the OEIS as A007154, *the free spectrum of Kleene algebras in the constant-free signature* (Berman [1]): dropping the constants $0, 1$ from the signature removes exactly two elements from each free algebra. Theorem 3.2 gives the same numbers a different meaning — the number of compatible spheres in the Kleene cube category — via the bijection boundaries $\leftrightarrow$ fiber-minimal elements. We do not know a direct structural link between the two interpretations beyond the count; both “remove two elements”, but different ones ($\{0, 1\}$ versus $\{\delta_\top, 1\}$).

Remark 3.4 (Boolean and Dedekind). For the Boolean cube category ($x \vee \neg x = 1$), the representation collapses to functions on the *consistent* points (all pairs $(0, 1)$ or $(1, 0)$): every point is a face point, no invisible points remain, and all fibers are singletons — boundaries determine fillers, as for the bare distributive (Dedekind) theory [8]. With Kalman’s theorem [7] — Boolean and Kleene are the only proper nontrivial subvarieties of De Morgan algebras — the table of §1 is therefore a *complete* classification of boundary-fiber behaviour over the subvariety lattice.

4. THE STRICT LAYER

We record how the structure theory of [9] specializes. Throughout, $r(g) = g \setminus A_g$ where A_g is the set of minimal invisible points of g ; by Theorem 3.2, $A_g \subseteq \{\perp, \top\}$, and indeed $A_g = \{\top\}$ iff $g = \delta_\top$, $A_g = \{\perp\}$ iff $g = 1$.

Proposition 4.1. *The Kleene cube category is not idempotent complete ($x \mapsto x \vee \neg x$ does not split; the proof of [9, Prop. 3.1] applies verbatim since $|\text{KL}_1| = 6$), hence not Eilenberg–Zilber; and its Karoubi envelope is the opposite of the category of finite projective Kleene algebras, characterized combinatorially by Bova–Cabrer [3, Thm. 12].*

Theorem 4.2. *Over the Kleene cube category:*

- (1) (defect formula) *for every g , $r(g)$ is the least element of the boundary fiber of g in one deletion step;*
- (2) (coherence dichotomy) *for every substitution f (inducing F on unmixed points — substitutions preserve unmixedness) and every g : $r(f^*g) = r(f^*r(g))$ iff $F(\text{Face}(U_m)) \cap A_g = \emptyset$;*
- (3) (polar exceptional locus) *consequently the incoherent pairs (f, g) have $g \in \{\delta_\top, 1\}$: for each substitution, at most two elements — independent of n — fail to transport coherently, in contrast with the positive-fraction exceptional loci of the De Morgan case [9].*

Proof. (1) and (2) are the proofs of [9, Thm. 4.1, Thm. 5.2] verbatim over the poset U_n , with Lemma 3.1 supplying the two ingredients used there (faces see exactly face points; a face point lies between comparable invisible points). Preservation of unmixedness by substitutions is the semantic fact that lower/upper approximations are computed uniformly: $F(v_{\text{low}}) = (t(v))_{\text{low}}$ for the three-valued evaluation t of the substitution. (3) is immediate from $A_g \subseteq \{\perp, \top\}$ and the identification of the two elements with nonempty defect. $\square$

Remark 4.3. Thus the Kleene strict layer is “Eilenberg–Zilber up to two poles”: the minimal-filler retraction is a strictly natural normal form away from $\{\delta_\top, 1\}$, and even there the deviation is a single toggled pole, corrected inside a two-element fiber by the median cylinder (whose Kleene instance is the same term as in [8]). It is instructive that the two exceptional elements are precisely the witnesses of the *failure of excluded middle* surviving the Kleene law: $\delta_\top = \bigwedge_i (x_i \wedge \neg x_i) \neq 0$ and its dual. The Kleene law tames, but cannot eliminate, the no-LEM cells — and they are exactly what separates the Kleene cube category from Eilenberg–Zilber behaviour.

5. COMPLETENESS: FIBER BEHAVIOURS ACROSS SIGNATURES

The trichotomy of the introduction classifies fiber behaviour over the subvarieties of De Morgan algebras. One may still ask whether *weaker signatures* — interval theories that drop the involution or the connections — realize intermediate behaviours. They do not.

Lemma 5.1. *An interval signature (a subset of $\{\wedge, \vee, \neg\}$ over the bounds $0, 1$) containing $\neg$ and at least one connection defines the other connection by De Morgan’s laws. Hence the signature classes are: $\emptyset$, $\{\wedge\}$, $\{\vee\}$, $\{\wedge, \vee\}$ (the Dedekind theory), $\{\neg\}$, and the full De Morgan signature.*

Theorem 5.2 (completeness). *Over every interval signature not containing both the involution and a connection, all boundary fibers of the free algebras are singletons. Consequently the realizable fiber behaviours of interval theories are exactly three: trivial (Dedekind, Boolean, and all weaker signatures), two polar fibers (Kleene), and Sperner-extremal (De Morgan).*

Proof. By Lemma 5.1 four signature classes remain below the full one. *Dedekind:* the free bounded distributive lattice on n generators is the lattice

of monotone functions on $\{0, 1\}^n$; every evaluation point has a first coordinate in $\{0, 1\}$, so an element is determined by its two faces $x_1 := 0, 1$ — fibers are singletons. (The same two-face decomposition handles the *Boolean* algebra of all functions, re-proving the first line of the table.) *Meet-only* (dually join-only): the free bounded $\wedge$ -semilattice consists of 0 and the conjunctions $\bigwedge_{i \in S} x_i$; the faces of a conjunction determine S (whether $i \in S$ is read off the $x_i := 0$ face), and distinguish 0 from 1. *Involution-only*: the free algebra is $\{0, 1, x_1, \neg x_1, \dots\}$, and the faces of $x_i, \neg x_i$ (constants 0, 1 in opposite positions) separate all elements. *Bare*: immediate. In each case no evaluation point is invisible to the faces; the nontrivial behaviours require literals to decouple ($\neg$, giving independent pairs) *and* to combine ($\wedge$ or $\vee$), and once both are available the algebra is a De Morgan algebra, where Kalman’s theorem [7] and the fiber theorems ([8], Theorem 3.2) leave exactly the three grades. $\square$

Remark 5.3. The moral sharpens the through-line of the introduction: *fiber freedom is born only from the interaction of the involution with a connection, and once born it is exhausted by Kalman’s three grades*. In particular no interval *theory* interpolates strictly between the Kleene and De Morgan rows — any graded interpolation must come from non-equational structure (restricted substitution classes, cf. the coherence analysis of [9]), not from the algebra.

6. MACHINE VERIFICATION

All counts and theorems were verified exhaustively for $n \leq 3$ (scripts accompany the artifact): $|\text{Mon}(U_n)| = 6, 84, 43918$ against the Berman–Mukaidono spectra; fiber censuses $\{1^2, 2^2\}$, $\{1^{80}, 2^2\}$, $\{1^{43914}, 2^2\}$ confirming Theorem 3.2 and the identity $B_{\text{KL}}(n) = |\text{KL}_n| - 2$; the defect formula on all 84 elements of KL_2 against brute-force fiber minima; and the coherence dichotomy on all of KL_2 for five generating substitutions — conditions of Theorem 4.2(2) coincide exactly, with incoherent loci of sizes 2, 2, 0, 2, 0 (both connections, reversal, diagonal, $x \mapsto x \wedge \neg x$), each consisting of the two polar elements. Assertions that substitutions preserve unmixedness and that defects stay within $\{\perp, \top\}$ were checked at every step. For Theorem 5.2, the singleton-fiber claims were enumerated as well: Dedekind $n \leq 3$ (6 and 20 elements), meet-only $n \leq 3$ (5 and 9 elements), involution-only $n = 2$ (6 elements) — all fibers singleton. At $n = 2$ the boundary counts of the Kleene and De Morgan theories coincide (82): the mixed points of Q_2 are the two non-polar diagonals, so the two face-point posets agree — a low-dimensional accident that disappears at $n = 3$ (43,916 versus 4,829,136 [8]).

7. OUTLOOK

Two questions remain. First, with the equational axis closed by Theorem 5.2, graded interpolations between the Kleene and De Morgan rows can only arise from *non-equational* data — wide subcategories of the cube

category (restricting which substitutions act) or filtered mixing constraints; there the varying quantity is not the fiber but the *coherence* behaviour of [9], and a classification over the generator classes of [4] seems within reach. Second, *the strict-weak interface*: in the De Morgan case the diagonal points simultaneously index fiber freedom, coherence defects, and the failure of idempotent splitting; the Kleene case shows the first two can be cut to a constant while the third persists ($x \vee \neg x$ still does not split). Whether the Karoubi envelope of the Kleene cube category — the finite projective Kleene algebras of [3] — admits a boundary-fiber theory of its own, and how the polar cells behave there, is open.

Artifact. Scripts and machine-checked companions are available in the repository at <https://github.com/r-shibusawa/cubical-strict-j> (archived at DOI 10.5281/zenodo.21405961).

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DAIICHI INSTITUTE OF TECHNOLOGY, JAPAN
 Email address: r-shibusawa@daiichi-koudai.ac.jp